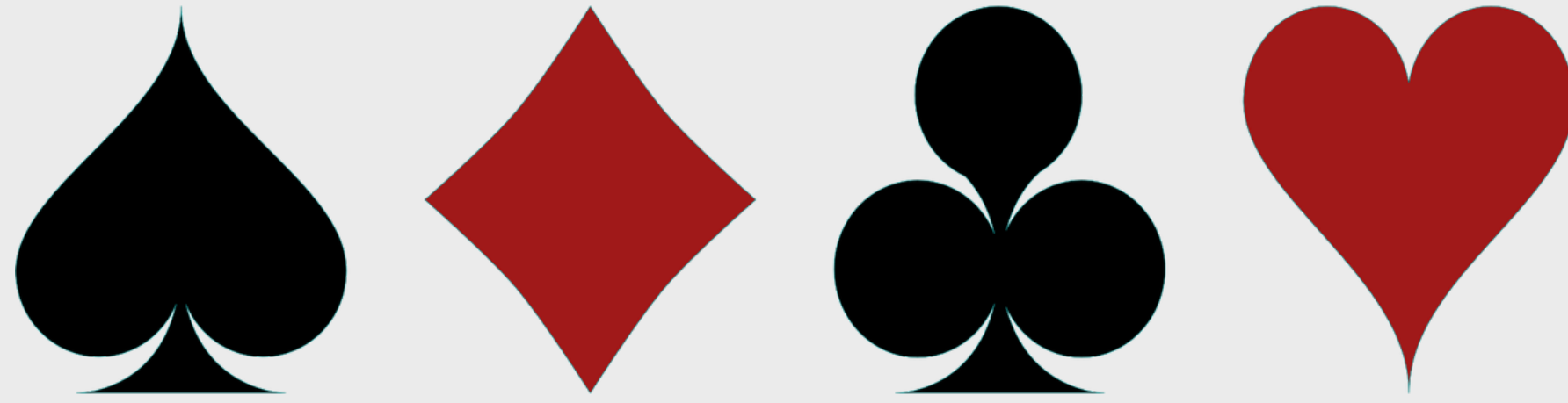




POKER ANALYTICS

**An analysis on how different factors in the game
affect the decision making of the player**

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OBJECTIVE

To study how poker players make decisions during a game and identify the factors that affect those decisions. The project analyzes factors such as hand strength, player position, betting patterns, pot odds, etc. It also aims to quantify hand strengths to understand how card quality influences actions like fold, call, or bet. Through poker analytics, the project tries to understand and model player strategy and decision-making.





FACTOR AFFECTING PLAYER'S DECISION

We will analyse the following independent factors and then use a combination of those to make the decision. We will add weights in each possible scenario to the factors in order to get a quantified final choice.

- Hand strength.
 - Resources (chips) and stakes.
 - Impression of the opponent's cards.
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FACTOR AFFECTING PLAYER'S DECISION(CONTD.)

01

Hand Strength (range based).

Once a player receives their cards, we quantify the odds of their hand being a winning hand. We could calculate a range of possibilities that this hand could end up in and the likelihood of that happening. Once we get the flop, turn and river we also calculate at each step the possible range of hands stronger than our hero's and the probability of the opponent attaining those hands.

FACTOR AFFECTING PLAYER'S DECISION(CONTD.)

02

Resources and stakes

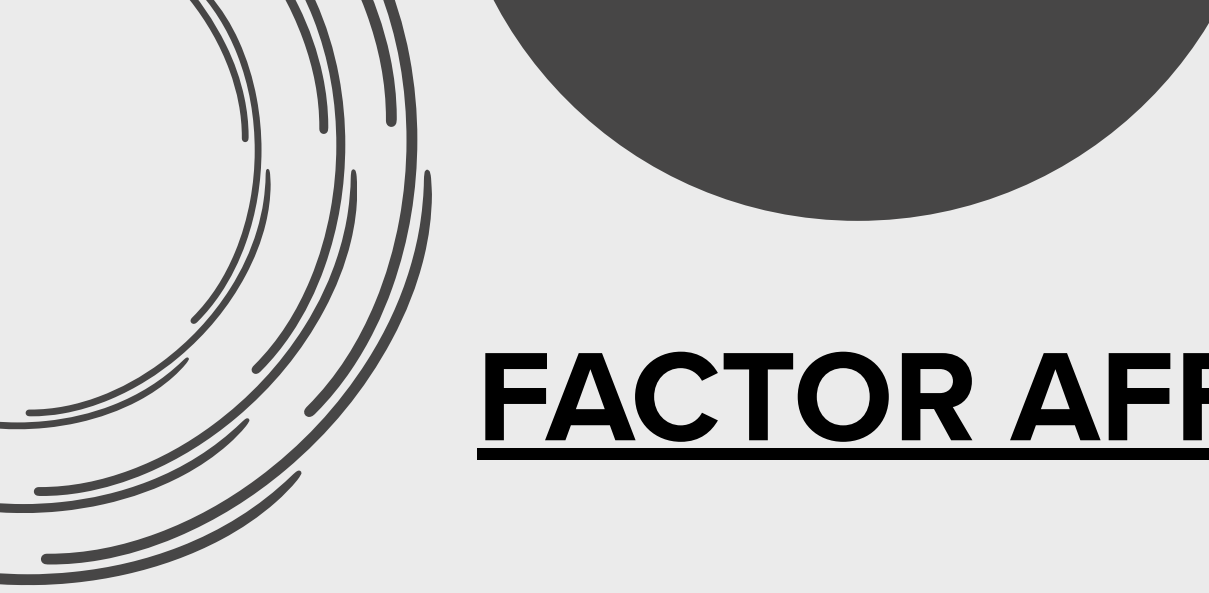
Stack-pot ratio(SPR):

This is the ratio of effective stack $\min(\text{our_chips}, \text{opp_chips})$ to pot_size . We set thresholds for SPR to add different weights to the factors at each play. We shall calculate SPR at flop.

03

Impression of opponent's hand

This is a complex factor which will be used to calculate a range of possible hands the opponent might possess based on their plays in the current game and their habits so far.



FACTOR AFFECTING PLAYER'S DECISION(CONTD.)

A combination of the mentioned factors shall be used to determine weights. We will have a threshold for stack-pot ratio (SPR), ϵ . If $SPR < \epsilon$, the weight for hand strength will be very high.

In the other scenario, the weights will be found using our dataset for the 2025 WSOP champion.



QUANTIFYING HAND STRENGTHS

The following are the hand rankings in a Texas Poker game, listed from strongest to weakest:
(SOURCE: [World Poker Tour](https://www.worldpoker.com/))

1. Royal Flush



The strongest hand in poker, a royal flush, includes an Ace, King, Queen, Jack, and 10, all in the same suit.

2. Straight Flush



A straight flush is a rare hand made up of any straight where all five cards share the same suit.

3. Four Of a Kind



Four of a kind, or "quads," is composed of four cards of the same value, accompanied by a side card of a different value.

6. Straight



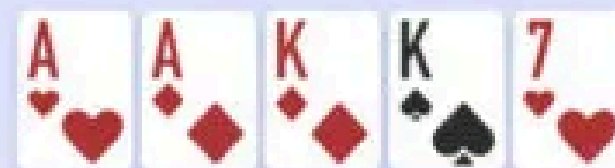
A straight consists of five consecutive cards in value, but not necessarily of the same suit.

7. Three of a Kind



Also known as "trips," three of a kind includes three cards of the same value, with two additional cards of different values.

8. Two Pair



Two pair is a hand containing two cards of the same value, another two cards of a different matching value, and one extra card.

4. Full House



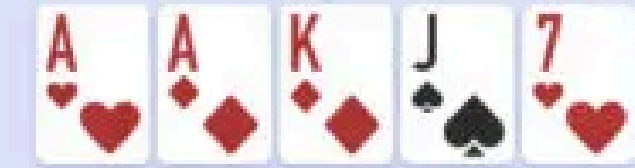
A full house consists of three cards of one rank and two cards of another.

5. Flush



A flush is a hand where all five cards are of the same suit.

9. Pair



One pair is a hand that has two cards of the same value, with the remaining three cards being of different values.

10. High Card



A high card hand features five cards that don't combine to form any of the above hands.





QUANTIFYING HAND STRENGTHS

To quantify the strength of our(2 hole) cards on basis of the board cards(pre-flop, flop, turn, river), we can use the **Effective Hand Strength algorithm(EHS)**.

(source: https://en.wikipedia.org/wiki/Effective_hand_strength_algorithm)

We define the two before proceeding:

1. **Hand Evaluator:** Computers are slow at reading strings, so convert cards into 32-bit integers. Use a pre-built Dictionary/Lookup Table. You input 5 cards, it instantly outputs a static Rank Number (eg: Royal Flush = 1).
 2. **Hand Strength(HS):** Hand Strength measures how your hand currently fares against a random opponent hand, ignoring future community cards. This will naturally yield a number between 0 and 1.
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QUANTIFYING HAND STRENGTHS

Algorithm:

1. Identify all known cards (your 2 hole cards + any community cards on the flop, turn, or river).
2. Identify the remaining unknown deck.
3. Iterate through every possible 2-card combination the opponent could hold from the unknown deck.
4. For each combination, use your Hand Evaluator to compare your best 5-card hand against their best 5-card hand.
5. Keep a tally of **W(Wins)**, **T(Ties)**, and **L(Losses)**.

The Formula:

$$HS = (W+T/2) / (W+T+L)$$

Note:

1. In a tie, the players split the pot 50/50. Therefore, a tie is mathematically worth half of a win.
2. On the River, there are no more cards to come, so your final score is simply HS. However, on the Flop and Turn, HS is not enough, because a weak hand (like a flush draw) has a low HS but high potential in future.

QUANTIFYING HAND STRENGTHS

- **PPOT(Positive Potential):** Probability that a hand currently losing will improve to win.
- **NPOT(Negative Potential):** Probability that a hand currently winning will be overtaken.
- **EHS(Effective Hand Strength):** Probability of winning after the final card is dealt.

Algorithm:

1. Consider the cases when the player is **AHEAD**(winning) and **BEHIND**(losing).
2. Since NPOT is the chance your opponent catches a lucky card to beat you, (1-NPOT) is the chance that you safely hold up. Therefore HS is when we are **currently AHEAD** and $HS \times (1 - NPOT)$ is the probability of **staying AHEAD**.
3. Likewise, (1-HS) is when we are **currently BEHIND** and $(1 - HS) \times PPOT$ is the probability of **improving to get AHEAD**.

$$\begin{aligned} EHS &= P(\text{AHEAD}) * P(\text{stays AHEAD}) + P(\text{BEHIND}) * P(\text{improves to get AHEAD}) \\ EHS &= HS \times (1 - NPOT) + (1 - HS) \times PPOT \end{aligned}$$



QUANTIFYING HAND STRENGTHS

Therefore, based on EHS a player may typically adjust their decision making as follows:

- **High EHS** : confidence and aggression (**Tend to Raise**)
 - **Medium EHS** : caution and pot control (**Tend to Call**)
 - **Low EHS** : defensive play or folding (**Tend to Fold**)
- 



THANK YOU

